

Endotracheal Intubation

Guideline Title: Endotracheal Intubation

Guideline Number: GUID-3203-PR020

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CCCXI. **PURPOSE:** To secure a patient airway due to actual or potential airway compromise due to depressed sensorium, who require ventilatory assistance, or whose combativeness threatens airway compromise.

CCCXII. **DEFINITIONS:**

- A. ETT – Endotracheal tube
- B. NRB – Nonrebreather Mask
- C. BVM – Bag valve Mask
- D. ETCO₂ – End-tidal Carbon Dioxide

CCCXIII. **DEPARTMENT GUIDELINE:**

A. **EQUIPMENT:**

- i. Appropriate personal protective equipment following current OH infection prevention guidelines.
- ii. Oxygen
- iii. 2 endotracheal tubes of appropriate size (cuffs checked; stylets available).
- iv. Video laryngoscopy equipment (e.g., C-MAC)
- v. Standard Laryngoscope handle and appropriate size blade (lights checked) as backup.
- vi. BVM unit with reservoir and appropriately sized mask and PEEP valve
- vii. Suction equipment with Yankaur tip attached.
- viii. 10ml syringe
- ix. Cardiac monitor for monitoring of capnography (ETCO₂), EKG, and vital signs.
- x. Tape or commercial device to secure tube
- xi. 100% NRB mask and/or Nasal cannula capable of delivering oxygen at 10-25 L/min
- xii. Broselow tape for pediatric patients
- xiii. Rescue equipment: Supraglottic airway devices (King LT-D)

- xiv. Gum elastic bougie
- xv. High Efficiency Particulate Air (HEPA) filter for suspected infectious patients.
- xvi. Magill forceps
- xvii. Medications see *GUID-G008 - Rapid Sequence Intubation (RSI) for Endotracheal (ET) Intubation*
- xviii. Surgical airway equipment see *GUID-PR007 - Cricothyrotomy – Surgical*
- xix. Stethoscope

B. INDICATIONS:

- i. Patient unable to protect his/her own airway.
 - 1. Loss of gag/cough reflex or inability to handle secretions
- ii. Patient is incapable of sustaining adequate intrinsic ventilatory effort.
 - 1. Apnea or signs of hypoxia
 - 2. Glasgow coma scale <8
 - 3. Airway obstruction due to edema, burns, epiglottitis
- iii. Patient will need airway management to facilitate safe transport
 - 1. Anticipated loss of control of airway due to edema, trauma, or stridor
- iv. Patient will need prolonged ventilatory support
 - 1. Need for controlled mechanical ventilation due to ventilator or oxygenation failure from sedation, narcosis, stroke, brain injury, spinal cord injury, nerve injury, motor neuron disease, muscular problems, chest wall injuries, pleural effusions, pneumothorax, hemothorax, bronchospasm, inhaled foreign object, acute lung injury, pulmonary edema, or anaphylaxis, hemodynamic instability, multisystem failure, metabolic issues.
- v. Persistent hypoxemia despite all efforts to improve oxygenation.

C. CONTRAINDICATIONS:

- i. Patients who are maintaining a patent airway
- ii. Anticipated difficulty ventilating patients after paralysis

D. PROCEDURE:

- i. Perform an assessment identifying the need for airway intervention.
- ii. Obtain a baseline neurological assessment.
- iii. Apply personal protective equipment
- iv. Prepare the patient
 - 1. Apply cardiac monitor, pulse ox, and noninvasive BP and obtain initial readings.
 - a. If patient is hypotensive, consider volume resuscitation with IV fluids or blood products per *GUID-M011 – Hypotension* and/or *GUID-PR025 - Emergency Blood Product Administration* prior to RSI attempt.

2. If patient is conscious, get a history and explain procedure, and reassure patient.
 3. Gather equipment above. Ensure all equipment is set up. Check and have back up devices ready including supraglottic devices and emergency cricothyrotomy kit.
 4. Establish IV access if patient requires medications to facilitate intubation.
 5. Assess anterior neck for possible surgical airway.
 6. Select appropriate tube size using Broselow tape for pediatric patients.
- v. Pre-oxygenate patient with high flow O₂ by NRB mask. If available, simultaneously use a standard nasal cannula and leave in place during procedure for passive oxygenation.
 1. Do not manually ventilate the patient using a BVM unless respiratory effort is ineffective, as this may cause gastric distention with vomiting and aspiration.
 2. If unable to maintain oxygen saturation >93%, suction the mouth and oropharynx to decrease risk of view obstruction and ventilator-associated pneumonia. Consider 2-person thumbs up bagging technique with PEEP valve. Squeeze BVM in synchrony with patient's intrinsic respiratory effort.
 - vi. Consider elevating the head of bed 30 degrees. Position the head in sniffing position if possible. If trauma is suspected, consider removing the cervical collar (if present) while providing in-line manual immobilization of the head and neck.
 - vii. *Video laryngoscopy should be utilized for initial attempt unless the equipment is unavailable/malfunctioning or a documented clinical reason (i.e. copious blood in the airway) for direct laryngoscopy is given.*
 - viii. Ensure proper functioning of all equipment including suction, oxygen, medications, and airways (appropriate sizes, cuff integrity, bougie setup).
 - ix. Once intubating conditions are optimized, administer medications if needed per *GUID-3203-G008 Rapid Sequence Induction for Endotracheal Intubation*
 - x. Introduce the laryngoscope into the mouth.
 1. If using video device, insert blade into mouth and walk down airway. When uvula identified lift to expose epiglottis and slowly advance to expose airway.
 2. If using standard laryngoscope, sweep tongue to the left and advance blade to the base of the epiglottis (vallecula). Lift laryngoscope upwards and outwards to expose the glottis.
 3. If using a straight blade, directly elevate the epiglottis.
 - xi. Visualize the vocal cords. Consider utilizing external laryngeal manipulation by having your teammate place their hand on the

tracheal cartilage and gently shifting it until the trachea is in better view

- xii. Consider gentle cricoid pressure is suspicious for a high risk of aspiration
- xiii. If you encounter a partial view utilize a gum elastic bougie per *GUID-PR011 – Gum Elastic Bougie*.
- xiv. Advance the tip of the endotracheal tube through the vocal cords. Once the cuff has passed through, stop advancing (approximately 2-3cm).
- xv. Secure the tube and inflate the cuff until there is no air leak.
- xvi. Confirm placement by placing an ETCO₂ detector on ET tube. Begin ventilations with BVM at 100% oxygen. Auscultate for air movement in stomach and bilateral breath sounds. Observe chest rise and condensation in the ETT.
 - 1. If tube placement is unable to be confirmed or there exists any doubt as to correct tube placement, immediately remove tube and oxygenate patient with a BVM to maintain SpO₂ >93%.
 - 2. If right main stem intubation is suspected (decreased or absent left sided breath sounds, asymmetrical chest wall movement with ventilation, decreasing SpO₂ despite good capnographic waveform), deflate the cuff, slightly pull back on the tube (1-2cm) and evaluate tube placement again, then re-inflate the cuff.
- xvii. Secure the tube with tape or commercial device.
- xviii. Note the depth in cm at the teeth or gums.
- xix. If unable to visualize vocal cords or pass ET tube after 20 seconds, stop the attempt, ventilate patient with 100% oxygen via BVM for 30-60 seconds, and re-attempt intubation.
- xx. If intubation fails after 2 attempts and you are unable to adequately ventilate the patient with a BVM, consider placing a King tube per *GUID-PR016 - King LTS-D* or other supraglottic device if available.
- xxi. If the King or other supraglottic device is unsuccessful and the patient is still unable to be ventilated with a BVM, consider cricothyrotomy per *GUID-PR007 – Cricothyrotomy – Surgical*.
- xxii. After intubation:
 - 1. Monitor vital signs, ETCO₂, and recheck placement frequently per *GUID-PR009 - End-Tidal CO₂ Monitoring*
 - 2. Consider placing a gastric tube for gastric decompression per *GUID-PR010 – Gastric Decompression*.
 - 3. Administer post-intubation sedation per *GUID-G008 - Rapid Sequence Intubation (RSI) for Endotracheal (ET) Intubation*, *GUID-G005 - Analgesia and Sedation Management*, or *GUID-G007 - Propofol Diprivan* (if ordered by sending facility).

4. Move patients to the transport ventilator per *GUID-PR024 - Ventilation with a Mechanical Ventilator*.
5. Verify tracheal tube position with any movement of the patient and upon turnover of care at receiving facility.

- E. **COMPLICATIONS:** Vomiting, cardiac dysrhythmias (bradycardia, PVC's, and V-fib), failed intubation, aspiration of stomach contents, pneumothorax, cervical vertebral injury, tear or trauma to bronchus, esophagus, mandible, teeth, laryngospasm, increased intracranial pressure.
- CCCXIV. **DOCUMENTATION:**
- A. EMS Charts: ETCO2 must be numerically documented with vital signs and a printed strip must be attached to the chart
- CCCXV. **REFERENCES:** None